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(54) **IN-SITU CAP FOR GERMANIUM
PHOTODETECTOR**

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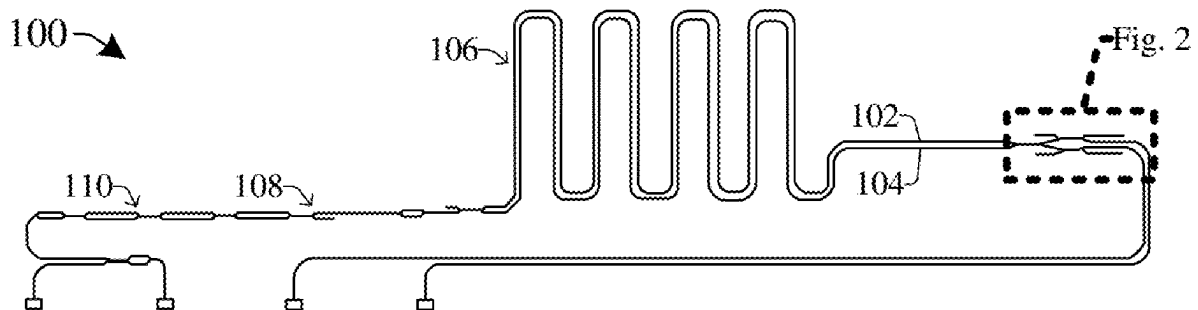
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division of application No. 17/197,353, filed on Mar.
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27, 2020.

(57)

ABSTRACT

Various embodiments of the present disclosure are directed towards an optoelectronic device. The device includes a substrate, and a germanium photodiode region extending into an upper surface of the substrate. The germanium photodiode region has a curved upper surface that extends past the upper surface of the substrate. A silicon cap overlies the curved upper surface of the germanium photodiode region. There is an absence of oxide between the curved upper surface of the germanium photodiode region and an upper surface of the silicon cap.



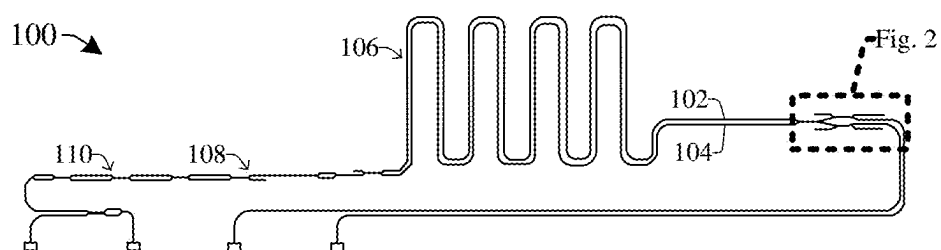


Fig. 1

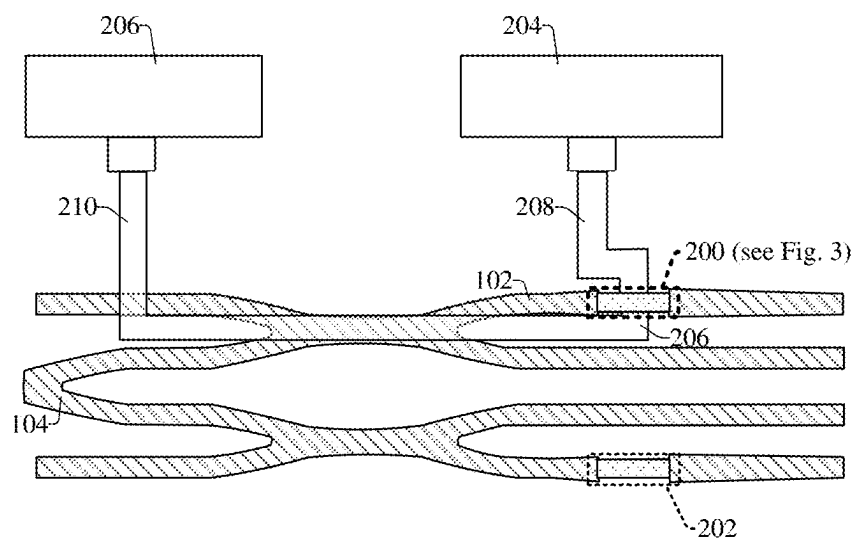


Fig. 2

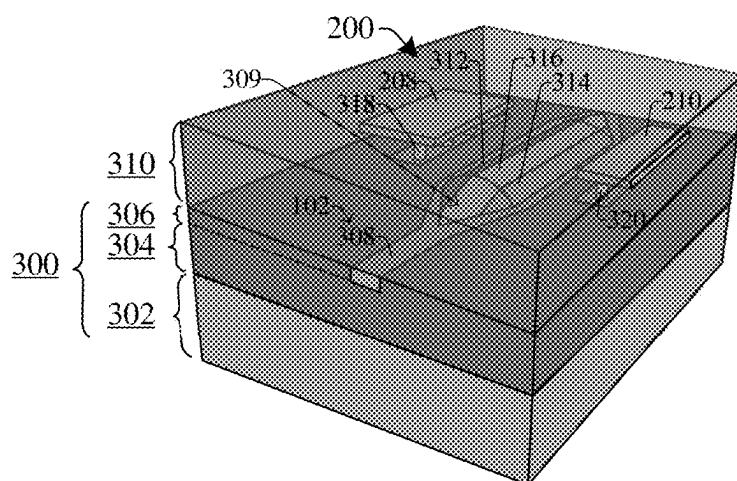


Fig. 3

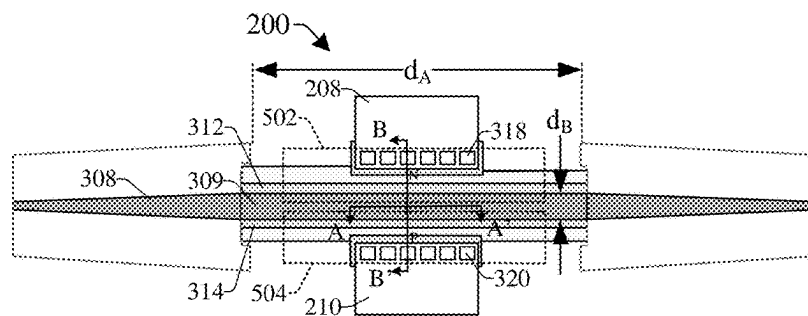


Fig. 4

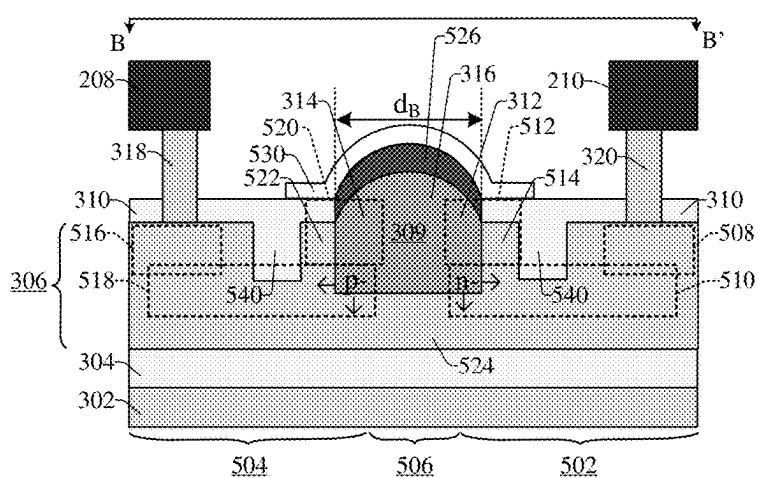


Fig. 5A

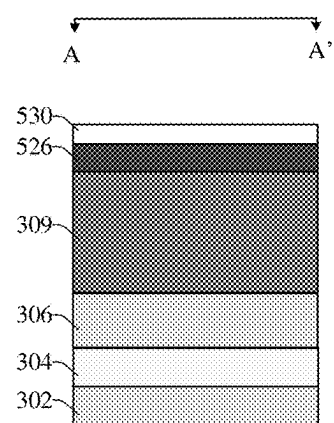


Fig. 5B

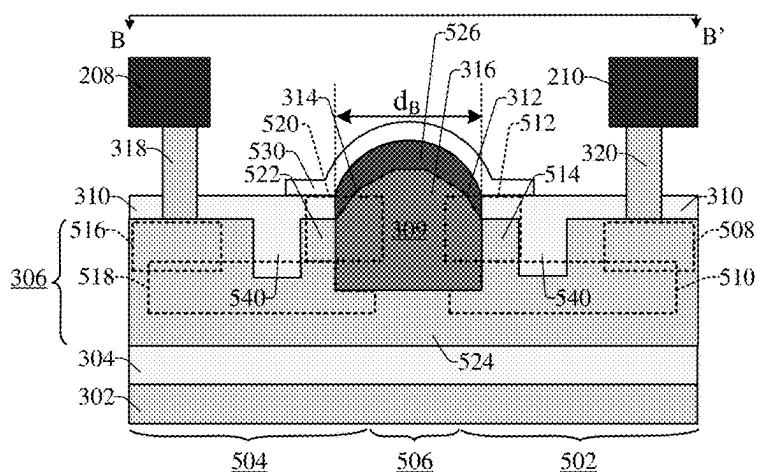


Fig. 6A

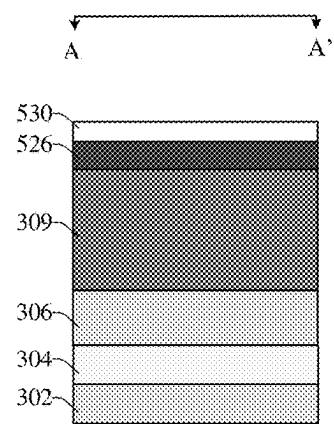


Fig. 6B

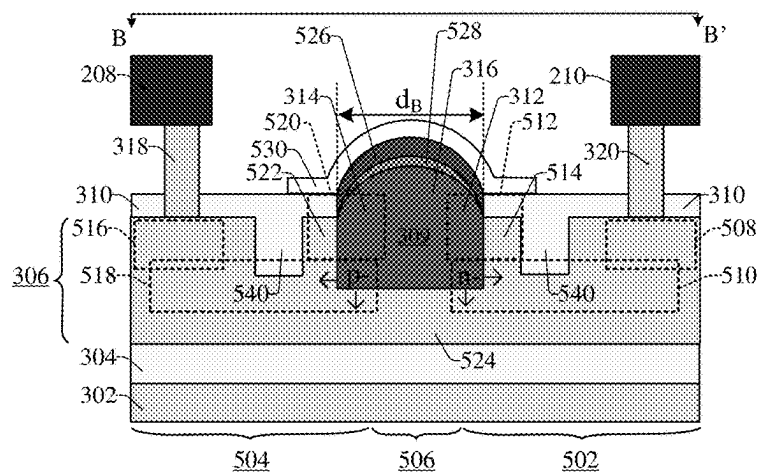


Fig. 7A

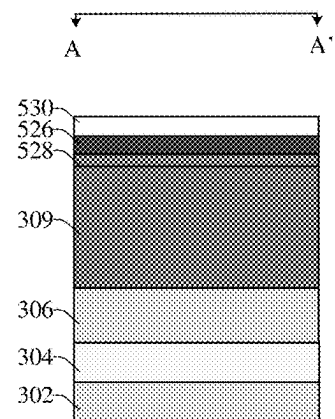


Fig. 7B

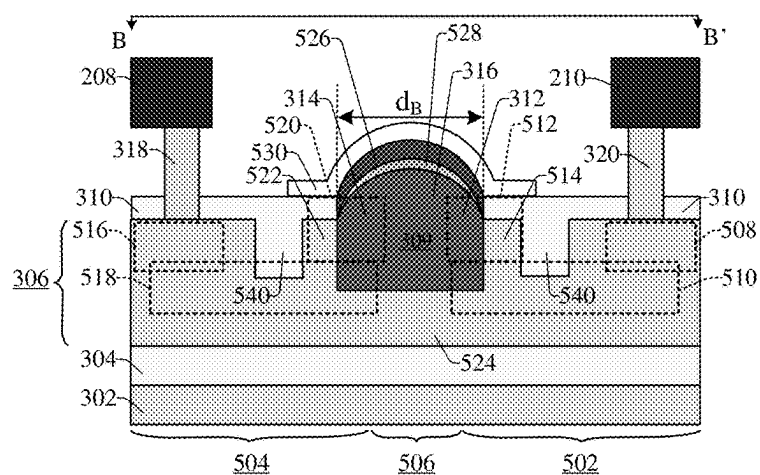


Fig. 8A

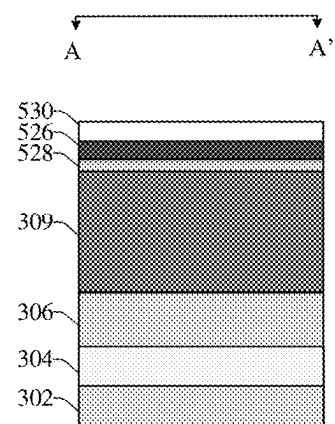


Fig. 8B

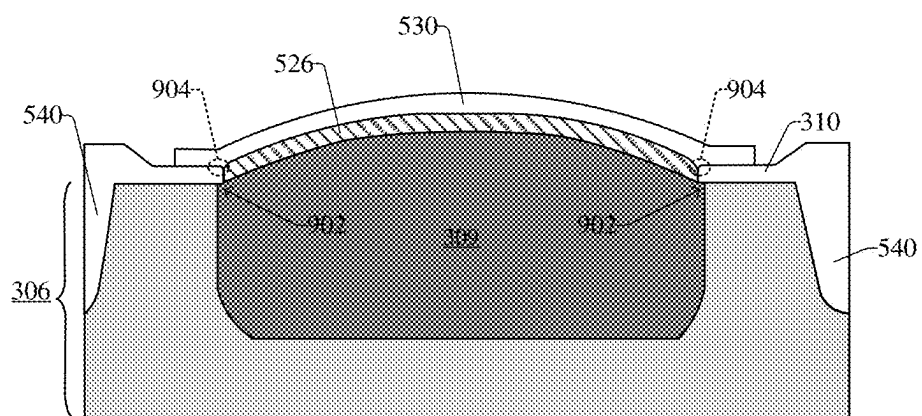


Fig. 9

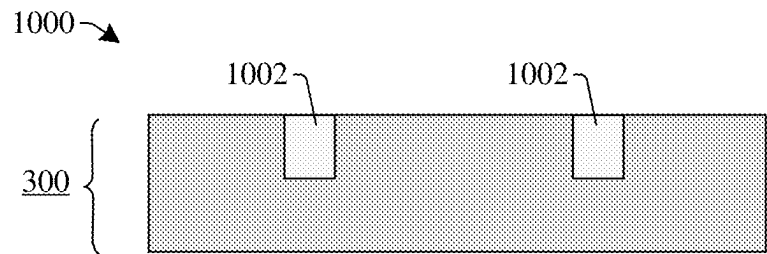


Fig. 10

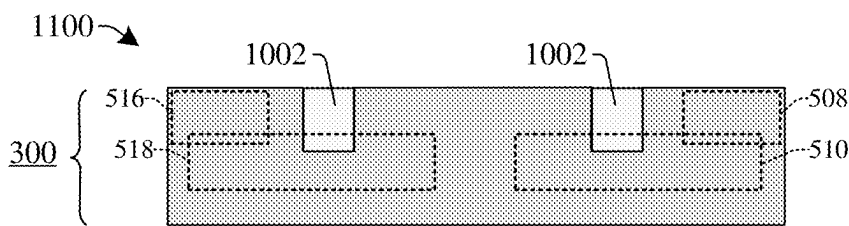


Fig. 11

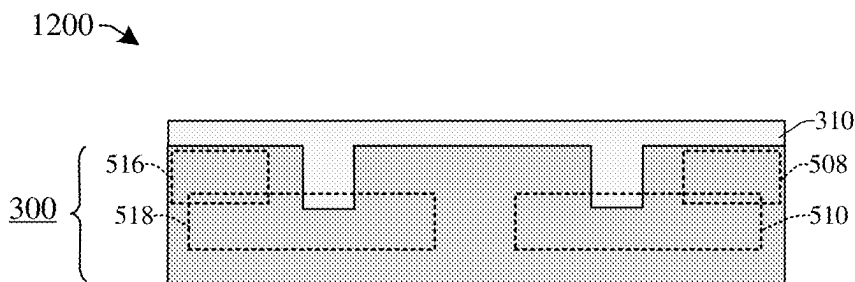


Fig. 12

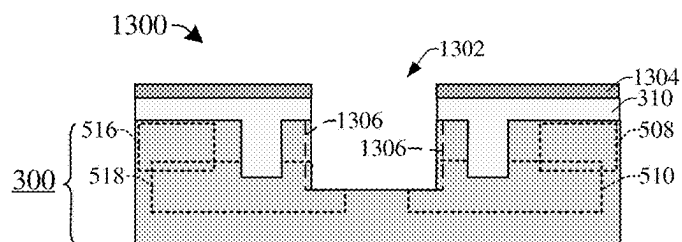


Fig. 13

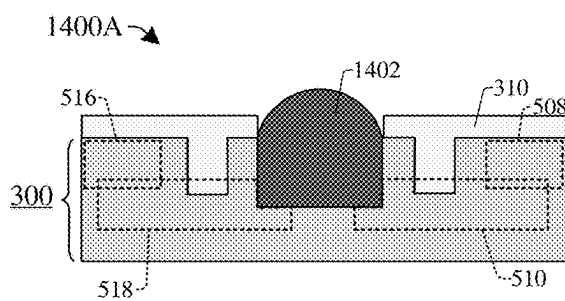


Fig. 14A

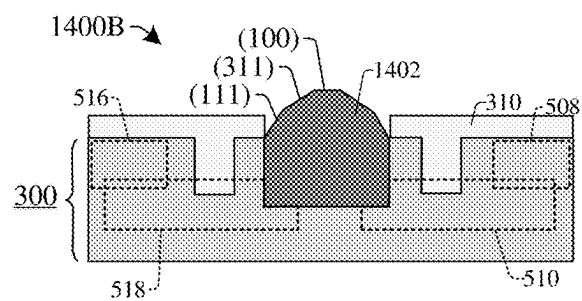


Fig. 14B

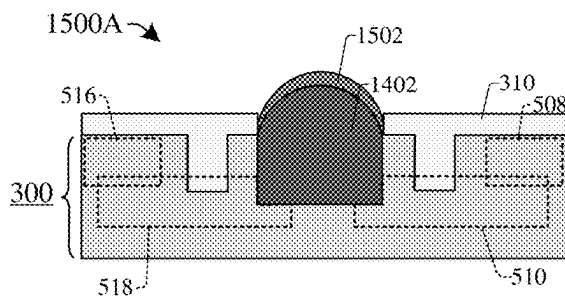


Fig. 15A

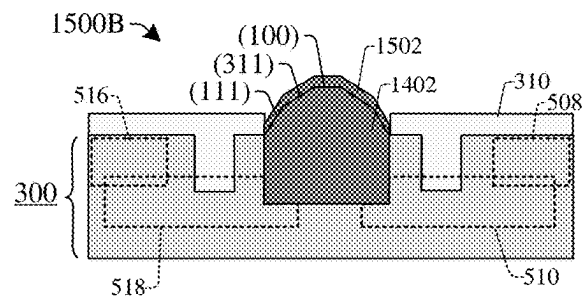


Fig. 15B

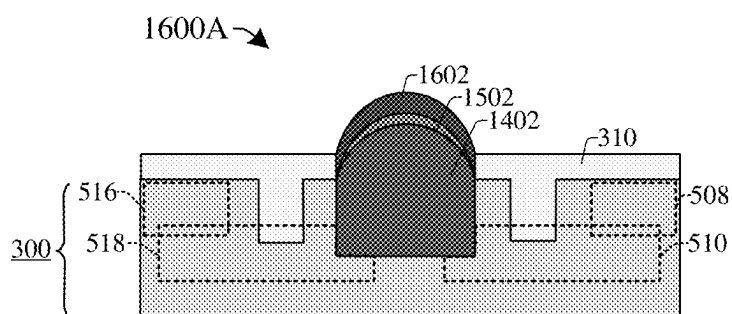


Fig. 16A

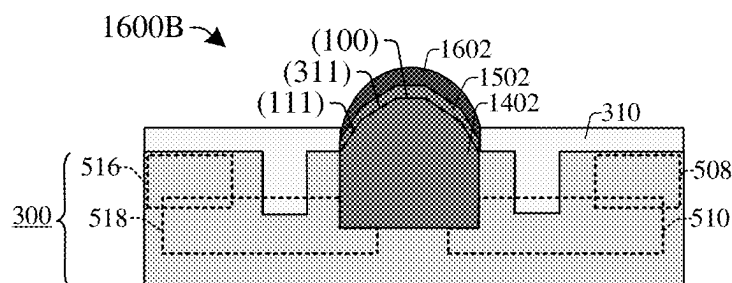


Fig. 16B

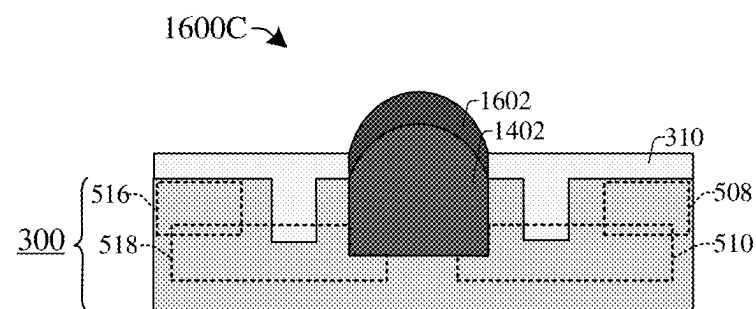


Fig. 16C

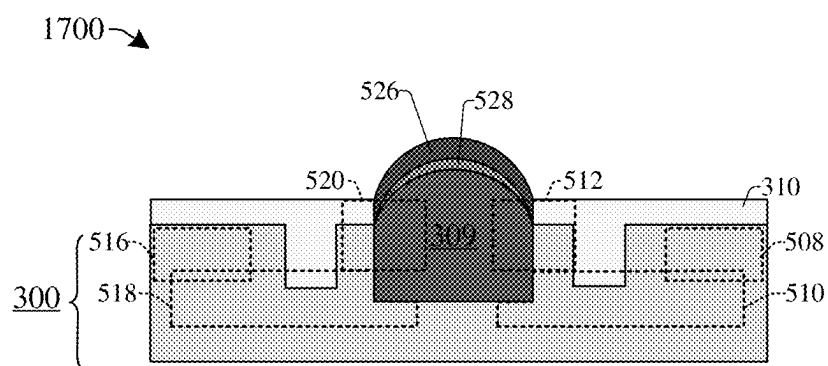


Fig. 17

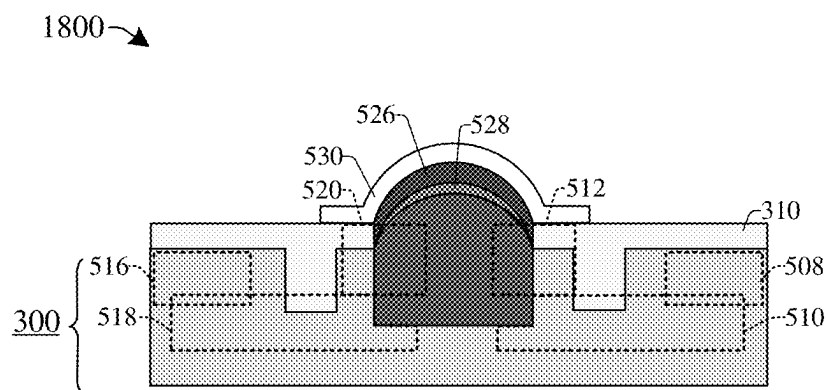


Fig. 18

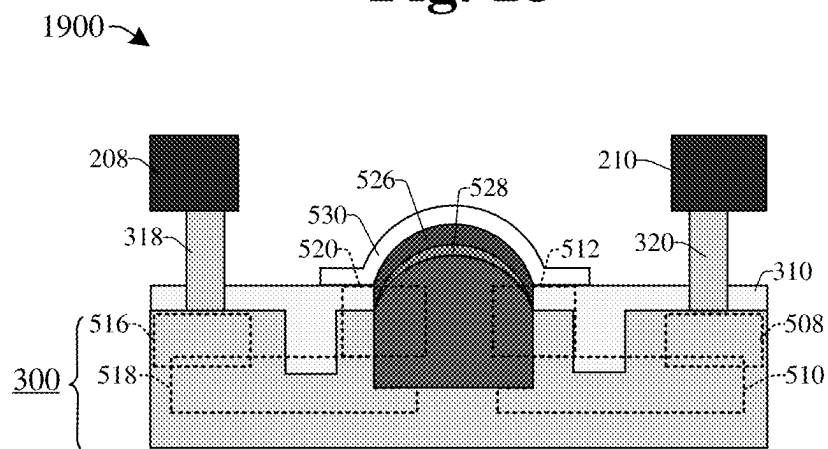
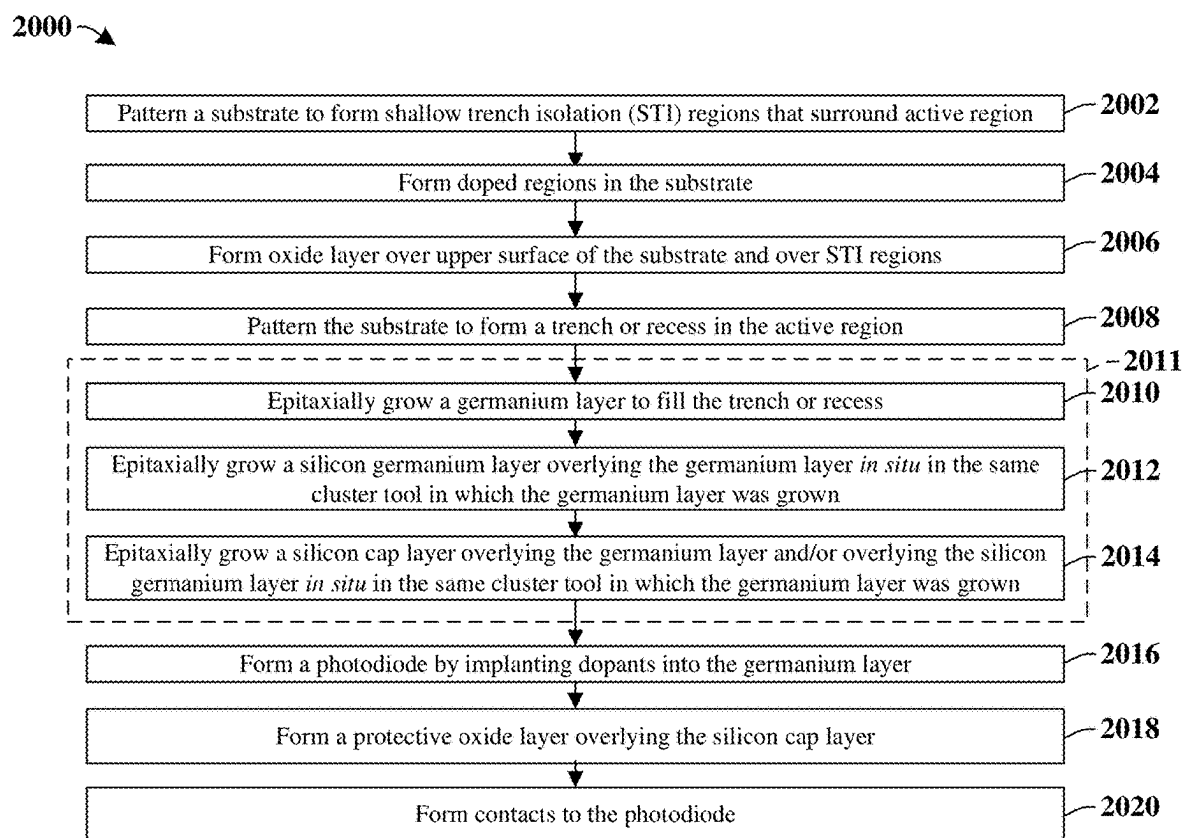


Fig. 19

**Fig. 20**

IN-SITU CAP FOR GERMANIUM PHOTODETECTOR

REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. application Ser. No. 18/592,715, filed on Mar. 1, 2024, which is a Divisional of U.S. application Ser. No. 17/197,353, filed on Mar. 10, 2021 (now U.S. Pat. No. 11,949,030, issued on Apr. 2, 2024), which claims the benefit of U.S. Provisional Application No. 63/106,019, filed on Oct. 27, 2020. The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] Silicon photonic devices can be made using existing semiconductor fabrication techniques, and because silicon is already used as the substrate for most integrated circuits, it is possible to create hybrid devices in which the optical and electronic components are integrated onto a single microchip. Consequently, silicon photonics is being actively researched by many electronics manufacturers, as well as by academic research groups, as a means for keeping on track with Moore's Law, by using optical interconnects to provide faster data transfer both between and within microchips.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates an example of an optoelectronic device in accordance with some embodiments.

[0005] FIG. 2 illustrates an inset view of a portion of the optoelectronic device consistent with FIG. 1 in accordance with some embodiments.

[0006] FIG. 3 illustrates a three dimensional view corresponding to a germanium photodiode consistent with some embodiments of the inset portion of FIG. 2.

[0007] FIG. 4 illustrates a top view of some embodiments of a germanium photodiode consistent with FIG. 3.

[0008] FIGS. 5A and 5B illustrate various cross-sectional views of some embodiments of the germanium photodiode of FIG. 4.

[0009] FIGS. 6A and 6B illustrate various cross-sectional views of some other embodiments of the germanium photodiode of FIG. 4.

[0010] FIGS. 7A and 7B illustrate various cross-sectional views of some embodiments of the germanium photodiode of FIG. 4.

[0011] FIGS. 8A and 8B illustrate various cross-sectional views of some embodiments of the germanium photodiode of FIG. 4.

[0012] FIG. 9 illustrates another cross-sectional view of a photodiode in accordance with some embodiments.

[0013] FIGS. 10-19 illustrate a series of cross-sectional views that collectively depict a method of manufacturing a photodiode in accordance with some embodiments.

[0014] FIG. 20 illustrates a block diagram of some embodiments of the method of FIGS. 10-19.

DETAILED DESCRIPTION

[0015] The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0016] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0017] Optical fibers and optical waveguides include a core, and a cladding or substrate at least partially surrounding the core. The core has a core refractive index n_{mi} and the cladding has a cladding refractive index n_{no} , whereby the core refractive index is higher than the cladding refractive index. Thus, when a light beam is coupled to the optical fiber or waveguide (e.g., when the light beam is directed into an end of the optical fiber or waveguide), the light beam is confined within the core by total internal reflection as the light beam propagates down the length of the optical fiber or waveguide.

[0018] In some cases, the core comprises silicon and the cladding includes silica or silicon dioxide. A photodiode, which can be formed in a germanium photodiode region located at a predetermined location along the length of the waveguide, can be present to measure or detect light at the predetermined location. Some aspects of the present disclosure lie in the appreciation that it is advantageous to prevent the uppermost surface of the germanium photodiode region from oxidizing, as such oxidation can lead to increased dark current for the final device. In alternative approaches, a silicon nitride cap is formed over an upper surface of the germanium photodiode region to decrease or prevent such oxidation. However, the silicon nitride cap is susceptible to voids during manufacture due in part to the hardness of the silicon nitride, whereby the voids can lead to oxidation of the underlying germanium and/or the underlying germanium being attacked by chemicals in the manufacturing process. Because of this, some aspects of the present disclosure provide a silicon cap over the germanium photodiode region to eliminate or decrease oxidation. The silicon cap is formed in-situ with the germanium photodiode region, thereby preventing oxidation of the upper surface of the germanium photodiode region. Further, because the silicon cap is less hard than silicon nitride, the silicon cap is less susceptible to

voids and tends to reduce dark current relative to approaches that use a silicon nitride cap. Lastly, in some embodiments, the in-situ process used to form the germanium photodiode region and the silicon cap over the germanium photodiode region can save a photomask compared to some alternative approaches.

[0019] With reference to FIG. 1, one can see an optoelectronic device 100 that includes a germanium photodiode with a silicon cap in accordance with some embodiments. It will be appreciated that FIG. 1's optoelectronic device 100 is merely an example, and optoelectronic devices in accordance with this disclosure can take any number of forms, including waveguides, splitters, modulators, demodulators, couplers, de-couplers, multiplexers, de-multiplexers, transmitters, receivers, and a host of other applications. The illustrated optoelectronic device 100 includes a series of optical paths, e.g., first optical path 102 and second optical path 104, included in a waveguide. At various locations, the optical paths may run in parallel with one another (see arrow 106), may branch apart from one another (see arrow 108), or may merge with one another (see arrow 110) to facilitate desired functionality—again FIG. 1 is just an example.

[0020] FIG. 2 illustrates a more detailed view of an inset portion of FIG. 1's optoelectronic device 100 including first optical path 102 and second optical path 104. A photodiode 200 is disposed at a first predetermined location on the first optical path 102, and another photodiode 202 is disposed at another predetermined location on the second optical path 104. The photodiode 200 is coupled to two conductive metal pads, such as bond pads. More particularly, a first metal line 208 couples a cathode of the photodiode 200 to the first metal pad 204, and a second metal line 210 couples an anode of the photodiode 200 to the second metal pad 206.

[0021] FIG. 3 illustrates a three-dimensional/perspective cutaway view of the portion of FIG. 2 corresponding to the photodiode 200. As can be appreciated from FIG. 3, the optical paths (e.g., first optical path 102) can be disposed on a substrate 300, such as a silicon-on-insulator (SOI) substrate or bulk silicon substrate. In the illustrated embodiment, the substrate 300 is an SOI substrate including a handle substrate 302 comprising monocrystalline silicon, an insulator layer 304 comprising silicon dioxide, and a device layer 306 comprising silicon. A germanium photodiode region 309 is also embedded in the substrate 300. Thus, the first optical path 102 includes a core 308 comprising silicon and the germanium photodiode region 309 extending along an axis of the first optical path. A cladding, which often comprises silica or silicon dioxide, surrounds the core 308 and the germanium photodiode region 309. Thus, the cladding includes an oxide layer 310 (e.g., silicon dioxide layer) disposed over an upper surface of the core 308 and around sidewalls of the core 308, and the insulator layer 304 is disposed under the bottom surface of the core 308. The photodiode 200, which can be disposed fully or partially within the germanium photodiode region 309 and which is coupled to an end of the core 308 of the first optical path 102, can measure or detect light at a predetermined location.

[0022] To establish the photodiode 200, the germanium photodiode region 309 includes an n-type germanium region 312 and a p-type germanium region 314, and in the illustrated embodiment also includes an intrinsic germanium region 316 separating the n-type germanium region 312 and p-type germanium region 314 to establish a p-i-n germanium photodiode. In other embodiments, the intrinsic germanium

region 316 could be omitted and the n-type germanium region 312 could directly contact the p-type germanium region 314 at a p-n junction. A first set of one or more contacts 318 extends downward from the first metal line 208 to an n-type silicon region ohmically coupled to the n-type germanium region 312 (e.g., to the cathode of the photodiode 200) to couple the first metal pad 206 to the cathode of the germanium photodiode 200. A second set of one or more contacts 320 extends downward from the second metal line 210 to a p-type silicon region ohmically coupled to the p-type germanium region 314 (e.g., to the anode of the photodiode) to couple the second metal pad 206 to the anode of the photodiode 200. The core 308 may widen or taper at ends of the germanium photodiode region 309 to surround outer edges of the germanium photodiode region 309, and the cladding surrounds edges of the widened silicon core to facilitate total internal reflection through the waveguide.

[0023] FIG. 4 shows a top view of the germanium photodiode 200 and is consistent with some embodiments of FIG. 3; and FIGS. 5A-5B which are now described concurrently with FIG. 4 illustrate cross-sectional views along the width and length of the germanium photodiode 200, respectively, as shown in FIG. 4's top view.

[0024] As shown in FIGS. 4 and 5A-5B, the germanium photodiode 200 includes n-type region 502, p-type region 504, and intrinsic region 506 separating the n-type region 502 and p-type region 504 to establish a p-i-n photodiode. In other embodiments, the intrinsic region 506 could be omitted and the n-type region 502 could directly contact the p-type region 504 at a p-n junction. The n-type region 502 can include an n-type contact region 508 directly under the contact 320, a lateral n-type region 510 extending under a shallow trench isolation (STI) structure 540, an n-type photodiode region 512 (which may include n-type germanium region 312 and n-type silicon region 514). The p-type region 504 can include a p-type contact region 516 directly under the contact 318, a lateral p-type region 518 extending under the STI structure 540, a p-type photodiode region 520 (which may include p-type germanium region 314 and p-type silicon region 522). Intrinsic germanium region 316 separates the n-type germanium region 312 and p-type germanium region 314, and intrinsic silicon region 524 separates the lateral n-type region 510 from the lateral p-type region 518, thereby establishing a p-i-n photodiode. In some embodiments, the length of the germanium photodiode region, d_A , ranges from approximately 5 micrometers to approximately 50 micrometers, and is 15 micrometers in some cases. In some embodiments, the width of the germanium photodiode region, d_B , ranges from approximately 1/50th of d_A to equal to d_A ; and is 0.5 micrometers in some embodiments.

[0025] When light passes along the first optical path within the core 308, at some point the light reaches the first predetermined location of the germanium photodiode 200. When a photon of sufficient energy strikes the germanium photodiode 200, it creates an electron-hole pair. This mechanism is also known as the inner photoelectric effect. If the absorption occurs in the junction's depletion region, or one diffusion length away from it, these carriers are swept from the junction by the built-in electric field of the depletion region. Thus holes (p-) move toward the p-type anode (and towards p-type contact region 516), and electrons (n-) move toward the n-type cathode (and towards n-type contact region 508), and a photocurrent is produced such that the

magnitude of the current is proportional to the intensity of the light. The total current through the germanium photodiode **200** is the sum of the dark current (current that is generated in the absence of light) and the photocurrent, so the dark current should be minimized to maximize the sensitivity of the device.

[0026] As can be seen in FIG. 5A, the germanium photodiode region **309** has a substantially flat or level lower surface, vertical or substantially vertical sidewalls, and a curved upper surface. A silicon cap **526** is disposed over the germanium photodiode region **309**, and there is an absence of oxidation between the curved upper surface of the germanium photodiode region **309** and the lower surface of the silicon cap **526**. This absence of oxidation is due to the fact that the silicon cap **526** is formed in-situ with the germanium photodiode region **309** within a cluster tool, thereby preventing oxidation of the upper surface of the germanium photodiode region **309**. Further, the silicon cap **526** is relatively resistant to formation of voids, and thus, prevents subsequent damage during manufacturing and thereby tends to reduce dark current relative to alternative approaches that use a silicon nitride cap. In some embodiments, the silicon cap **526** has a thickness ranging from 100 angstroms to 1000 angstroms. The silicon cap **526** has a lower surface whose outer corners meet upper surface of the device layer **306**, and has an upper surface whose outer corners meet upper surface of the oxide layer **310**. As illustrated in FIG. 5B, in some embodiments the silicon cap **526** extends continuously over the germanium photodiode region and has a substantially uniform thickness along the length of the germanium photodiode region. Thus, a curved lower surface of the silicon cap is congruous with the curved upper surface of the germanium photodiode region.

[0027] A protective oxide layer **530** is disposed over the silicon cap **526**, and has outer edges that extend outwards past outer sidewalls of the germanium photodiode region **309** and that extend laterally over the upper surface of oxide layer **310**. In some embodiments, the protective oxide layer **530** is a low-k dielectric, but the protective oxide layer **530** can also be silicon dioxide, silicon nitride, silicon oxynitride, or another suitable dielectric material.

[0028] FIGS. 6A-6B shows an alternative embodiment where the germanium photodiode region **309** has a substantially rounded upper surface that includes a (111) facet, a (311) facet, and a (100) facet that collectively approximate a rounded surface. In some cases, the (111) facet and (311) facet appear due to the inner sidewalls of the oxide layer **310** constraining the epitaxial and selective growth of the germanium photodiode region **309**. Because the growth rate of the (100) facet is higher than (311) and (111) facets; and the growth rate of the (311) facet is higher than that of the (111) facet, the selectively germanium grown gives rise to the generally rounded surface of FIG. 14B.

[0029] FIGS. 7A-7B illustrates another embodiment where the germanium photodiode region **309** has a substantially flat or level lower surface, vertical or substantially vertical sidewalls, and a curved upper surface. A silicon cap **526** is again disposed over the germanium photodiode region **309**, and there is an absence of oxidation between the curved upper surface of the germanium photodiode region **309** and the lower surface of the silicon cap **526**. In addition, an interface layer **528** comprising silicon germanium is disposed between the curved upper surface of the germanium photodiode region **309** and the lower surface of the

silicon cap **526**. In some embodiments, the silicon germanium of the interface layer **528** has a thickness ranging from approximately 3% of the thickness of the silicon cap **526** to approximately 50% of the thickness of the silicon cap **526**. In some embodiments, the thickness of the silicon germanium ranges from 30 angstroms to 50 angstroms. In some alternative embodiments, the thickness of the silicon germanium is more than 0 angstrom and less than 30 angstroms. Further, the % of germanium by atomic ratio in the interface layer **528** is greater than 0% and less than or equal to 50% in some embodiments. In some embodiments, the interface layer **528** has a substantially constant atomic ratio of silicon to germanium throughout the entire thickness of the interface layer—for example the atomic ratio can be 1:1 for silicon:germanium from the upper surface of the germanium photodiode region **309** to the lower surface of the silicon cap **526**.

[0030] FIGS. 8A-8B illustrate another embodiment where the interface layer **528** comprises silicon germanium. Compared to FIGS. 7A-7B where the interface layer **528** has a substantially constant atomic ratio of silicon to germanium throughout its thickness, the atomic ratio of silicon to germanium in FIGS. 8A-8B is graded over the thickness of interface layer **528**. Thus, in FIGS. 8A-8B, the percentage of silicon by atomic ratio ranges from approximately 0% near the bottom to about 100% to the top, and can exhibit a linearly increasing percentage of silicon by atomic ratio (and percentage of germanium can correspondingly decrease) over this thickness. In other embodiments, the percentage of silicon by atomic ratio can increase (and percentage of germanium can correspondingly decrease) according to a quadratic function that varies continuously over the thickness, an exponential function that varies continuously over the thickness, or another continuous function; this is because continuous functions tend to control lattice strain and provide better crystalline structure than discrete, step-like changes in composition, which are also possible. Further, in some embodiments, the percentage of silicon germanium by atomic ratio ranges from approximately 100% near the bottom to about 0% to the top, and exhibits an increasing percentage of silicon by atomic ratio over this thickness. In some cases, the thickness of the interface layer **528** is approximately equal to that of the silicon cap **526**, however, the thickness of the interface layer **528** can also be greater than that of the silicon cap **526** or less than that of the silicon cap **526**. In some embodiments, the thickness of the interface layer **528** ranges from 150 angstroms to 1000 angstroms.

[0031] FIG. 9 shows another example illustrating some details of how the germanium photodiode region **309**, oxide layer **310**, and silicon cap **526** meet one another in embodiments. As shown in FIG. 9, in some embodiments, peripheral regions of the germanium photodiode region **309** can undercut the inner sidewalls of the oxide layer **310** (see 902). This is due to the process by which the germanium photodiode region **309** is formed, whereby a trench is formed in the substrate through etching that slightly undercuts the oxide layer **310**, and the germanium photodiode region **309** is then epitaxially and selectively grown within the trench. An upper surface of the oxide layer **310** and inner sidewalls can also have rounded corners **904** due to the etch. A central region of the germanium photodiode region **309** has an upper surface that is raised above an upper surface of the oxide layer **310**, and the silicon cap **526** is disposed over the

germanium photodiode region **309**. In some embodiments, the silicon cap **526** has a thickness that ranges from 75% to 125% of the thickness of the oxide layer **310**, and has the same thickness (plus or minus 5%) as the oxide layer **310** in some cases. A lower surface of the central region of the silicon cap **526** is raised above the upper surface of the oxide layer **310**. The protective oxide layer **530** is disposed over the silicon cap, and has outer edges that are approximately 50% between an outer edge of the germanium photodiode region and an inner edge of the STI structure **540**.

[0032] With reference to FIGS. **10-18**, a series of cross-sectional views **1000-1800** of some embodiments of a method for forming an optoelectronic device is provided in which a silicon cap layer is disposed over a germanium photodiode to reduce dark current.

[0033] As illustrated by the cross-sectional view **1000** of FIG. **10**, a substrate **300** is provided. The substrate **300** may, for example, be a bulk substrate of monocrystalline silicon or some other suitable type of substrate, such as an SOI substrate, among others. In some embodiments, the substrate **300** is un-doped. In other embodiments, the substrate **300** is doped with P-type or N-type dopants. The substrate **300** is patterned to form shallow trench isolation (STI) regions that extend into an upper surface of the substrate. The patterning to form the STI regions can comprise: 1) depositing a hard mask layer over the substrate **300**; 2) patterning the hard mask layer by a photolithography/etching process; and 3) performing an etch into the substrate **300** with the hard mask layer in place. The hard mask layer may, for example, be or comprise silicon nitride, un-doped silicate glass (USG) and/or some other suitable dielectric(s). The trench is the filled in with an insulating material, such as silicon dioxide or a low-k dielectric material for example. Optionally, a chemical mechanical planarization (CMP) operation is carried out to planarize an upper surface of the STI regions with the upper surface of the substrate **300**.

[0034] As illustrated by the cross-sectional view **1100** of FIG. **11**, p-type and n-type regions are formed in the substrate **300**. In some embodiments, the forming comprises: selectively performing ion implantation into the substrate **300** to form the p-type contact region **516**; selectively performing ion implantation into the substrate **300** to form the lateral p-type region **518**; selectively performing ion implantation into the substrate **300** to form the n-type contact region **508**; and selectively performing ion implantation into the substrate **300** to form the lateral n-type region **510**.

[0035] As illustrated by the cross-sectional view **1200** of FIG. **12**, an oxide layer **310** is then formed over the upper surface of the substrate **300**. The oxide layer **310** can be formed by a thermal oxidation, a spin on process, an atomic layer process, or a plasma deposition process, among others. In some embodiments, the oxide layer is a silicon dioxide layer, but can also be a low-k dielectric or another suitable dielectric.

[0036] As illustrated in cross-sectional view **1300**, the substrate **300** is then patterned to form a recess or trench **1302**. The patterning to form the trench **1302** comprise: depositing a hard mask layer **1304** over the substrate **300**; patterning the hard mask layer **1304** by a photolithography/etching process; and performing an etch into the substrate **300** with the hard mask layer **1304** in place. The hard mask layer **1304** may, for example, be or comprise silicon nitride, un-doped silicate glass (USG) and/or some other suitable

dielectric(s). In alternative embodiments, the patterning to form the trench **1302** is performed by some other suitable patterning process. In some embodiments, the etch may undercut the inner sidewalls of the oxide layer **310** (see lines **1306**) so the inner edges of the oxide layer **310** overhang the outermost edges of the trench **1302**.

[0037] As illustrated by the cross-sectional views **1400A-1400B** of FIGS. **14A-14B**, respectively, a germanium region **1402** is epitaxially grown filling the trench **1302** (see, e.g., FIG. **13**). Because the germanium region **1402** is epitaxially grown, the germanium region **1402** is selective to grow only on the silicon and grows outward from exposed surfaces of the substrate **300** in the trench **1302**. Further, because the hard mask layer **1304** covers the substrate **300** outside the trench **1302**, the germanium region **1402** is wholly or mostly localized to the trench **1302**. Other suitable materials are, however, amenable in alternative embodiments. Further, in some embodiments, the germanium region **1402** has a smaller bandgap than the substrate **300**. In some embodiments, the germanium region **1402** can have a generally rounded upper surface, which in some cases exhibits an absence of sharp corners or that includes several planar facets that collectively approximate a rounded surface. For example, FIG. **14A** shows a germanium region **1402** with a rounded upper surface that exhibits an absence of sharp corners, while FIG. **14B** shows a germanium region **1402** with a **(111)** facet, a **(311)** facet, and a **(100)** facet that collectively approximate a rounded surface. In some cases, the **(111)** facet and **(311)** facet appear due to the inner sidewalls of the oxide layer **310** that define an uppermost portion of trench **1302** and the fact the germanium grows selectively on the substrate **300** but not this oxide layer **310**. Because the growth rate of the **(100)** facet is higher than **(311)** and **(111)** facets; and the growth rate of the **(311)** facet is higher than that of the **(111)** facet, the selectively germanium grown gives rise to the generally rounded surface of FIG. **14B**. Whether the upper surface is rounded and exhibits an absence of sharp corners as in FIG. **14A** or includes several planar facets as in FIG. **14B**, the generally rounded upper surface can ultimately help to facilitate total internal reflection of light within the germanium photodiode region.

[0038] As illustrated by the cross-sectional views **1500A-1500B** of FIGS. **15A-15B** and **1600A-1600C** of FIGS. **16A-16C**, a cap layer is epitaxially grown on and covers the generally rounded upper surface of the germanium region **1402**. The cap layer may include a silicon cap layer **1602** (see FIGS. **16A-16C**) over the upper rounded surface of the germanium region **1402**, and may optionally include an interface layer **528** (see FIGS. **15A-15B**) between the silicon cap layer **1602** and the germanium region **1402**. Further, the cap layer is epitaxially grown, such that the cap layer grows on the germanium region **1402** but not the hard mask layer **1104**. As such, the cap layer is localized to the germanium region **1402** by a self-aligned process, and can save a photomask compared to some other approaches. Further, the cap layer is grown in-situ with the germanium region **1402** (meaning the cap layer is formed in the same cluster tool, or in even in the same chamber within a cluster tool, as the germanium region **1402** without exposing the substrate and germanium region to an ambient environment external to the cluster tool), such that the substrate and germanium region is well protected from the ambient environment. This in-situ processing helps prevent oxidation of the rounded upper surface of the germanium region **1402**, whereby such oxi-

dation would otherwise increase dark current in the resulting germanium photodiode. Therefore, this in-situ processing of the germanium region **1402**, optional silicon germanium interface layer **1502**, and silicon cap layer **1602** results in an absence of oxidation between the rounded upper surface of the germanium region **1402** and the uppermost surface of the silicon cap layer **1602**. In some regards, this in-situ processing has showed a significant reduction in dark current compared to other approaches that do not use this in-situ processing. For example, in some embodiments, a germanium photodiode with a silicon cap formed by in-situ processing, when having an applied voltage of -2 volts, an input power before the photodiode of 0.2 milliwatts (mW), an 3 dB bandwidth of 60 GHz, and an output current of 214 microamps, had a dark current of less than 20 nanoamps (nA), which is a reduction in dark current of more than 20 times compared to other approaches. This reduction in dark current is believed to arise because the lack of oxidation over the upper surface of the germanium region **1402** provides less interface charge than if the oxidation were present, and thus provides a significant improvement over other approaches.

[**0039**] More particularly in FIG. **15A**, a silicon germanium interface layer **1502** is formed on the rounded upper surface of the germanium region **1402**. The silicon germanium interface layer **1502** can be formed by an epitaxial growth process in situ with the germanium region **1402**. In FIG. **15A**, the silicon germanium interface layer **1502** can have a constant atomic ratio over its entire thickness, or can have an atomic ratio of silicon to germanium that changes over its thickness. In FIG. **15B**, the silicon germanium interface layer **1502** has facets that conform to underlying facets in the germanium region **1402**; and can have a constant atomic ratio over its entire thickness, or can have an atomic ratio of silicon to germanium that changes over its thickness. In some embodiments, the silicon germanium interface layer **1502** may be formed by introducing silicon germanium precursors in to the chamber to grow silicon germanium on the outer surface of the germanium region **1402**, and has a thickness ranges from 30 angstroms to 50 angstroms. In some alternative embodiments, the silicon germanium interface layer **1502** may be formed by introducing silicon precursors to react with germanium of the germanium region **1402**, and has a thickness more than 0 angstroms and less than 30 angstroms.

[**0040**] In FIGS. **16A-16C**, a silicon cap layer **1602** is formed over the germanium region **1402**. Again, the silicon cap layer **1602** is formed epitaxially and selectively, and in situ with the germanium region **1402** and the silicon germanium interface layer **1502** (if present). More particularly, in FIGS. **16A-16B**, which follow from FIGS. **15A-15B**, respectively, the silicon cap layer **1602** is formed over the silicon germanium interface layer **1502**. In FIG. **16C**, the silicon cap layer **1602** is formed directly over the germanium region **1402** with no intervening silicon germanium interface layer. The silicon cap layer **1602** protects the germanium region **1402** from damage during subsequent processing. For example, subsequent wet cleaning processes may use acids that have high etch rates for the germanium region **1402** but low etch rates for the silicon cap layer **1602**. As such, the germanium region **1402** would undergo significant crystalline damage and/or erosion but for the silicon cap layer **1602**. By preventing erosion and/or crystalline damage, leakage current is reduced.

[**0041**] As illustrated by the cross-sectional view **1700** of FIG. **17** (which follows from FIG. **16A**, but could also follow from FIGS. **16B-16C**), the germanium region **1402** and the silicon cap **526** are doped to form a germanium photodiode region **309**. Germanium photodiode region **309** comprises a p-i-n photodiode in the illustrated embodiment, but in other embodiments can comprise a PN photodiode. The p-i-n photodiode comprises a p-type photodiode region **520** and an N-type photodiode region **512**. To perform this doping, the substrate is removed from the cluster tool in which the germanium region **1402** and cap layer are formed, and is typically exposed to the ambient environment of the fab in being transferred to an ion implantation tool where the doping is performed. Thus, the top of the silicon cap layer **1602** may be oxidized at this point, but the underlying upper rounded surface of the germanium photodiode region **309** remains oxidation free to keep dark current at reduced levels.

[**0042**] As illustrated by the cross-sectional view **1800** of FIG. **18**, a protective oxide layer **530** is formed directly on the silicon cap **526** and over the inner edges of the oxide layer **310** nearest the germanium photodiode region **309**. In some embodiments, the protective oxide layer **530** comprises silicon oxide, and/or is a high k dielectric material with a dielectric constant in excess of 3.9 , 10 , or some other suitable value. The deposition may, for example, be performed by atomic layer deposition (ALD), vapor deposition, or some other suitable deposition process. In some embodiments, the protective oxide layer **530** is a metal oxide. Other suitable materials are, however, amenable. In some embodiments, the protective oxide layer **530** is or comprises aluminum oxide (e.g., Al_2O_3), titanium oxide (e.g., TiO_2), tantalum oxide (e.g., Ta_2O_5), hafnium oxide (HfO_2), zirconium oxide (e.g., ZrO_2), magnesium oxide (e.g., MgO), some other suitable high k dielectric(s), or any combination of the foregoing. In some embodiments, the protective oxide layer **530** has a thickness of about 1 - 10 nanometers, about 1 - 5 nanometers, about 5 - 10 nanometers, or some other suitable value.

[**0043**] As illustrated by the cross-sectional view **1900** of FIG. **19**, contacts **320**, **318** are formed to connect the contact regions **508** and **516**, respectively, to metal lines **210**, **208**, respectively. In some embodiments, the metal contacts and metal lines comprise copper, aluminum, nickel, lead, and/or tungsten, among other metals.

[**0044**] FIG. **20** illustrates some embodiments of methods **2000** for manufacturing an optoelectronic device in accordance with some embodiments. While examples of FIG. **20** are mentioned with reference to the preceding cross-sectional views of FIGS. **10-19**, it will be appreciated that the structures shown in FIGS. **10-19** are not limited to the methods **2000** but rather may stand alone separate of the methods. Further, while methods **2000** are described as a series of acts, it will be appreciated that the order of the acts (and/or portions of those acts) may be altered in other embodiments. For example, although FIG. **20** illustrates that a photodetector is formed by implanting dopants into the germanium layer in act **2016**, the dopants could also be implanted after **2018** or before **2014** in some embodiments, and other doped regions in **2004** could also be performed at other times in the method **2000**. Further still, while FIG. **20** illustrates a specific series of acts, some acts that are illustrated and/or described may be omitted in other embodi-

ments. Further, additional acts that are not illustrated and/or described in FIG. 20 may be included in other embodiments.

[0045] At act 2002, a substrate is patterned to form shallow trench isolation (STI) regions that surround an active region. In some embodiments, act 2020 can be consistent with FIG. 10.

[0046] At act 2004, doped regions are formed in the substrate. In some embodiments, act 2004 can be consistent with FIG. 11.

[0047] At act 2006, an oxide layer is formed over an upper surface of the substrate and over the STI regions. In some embodiments, act 2006 can be consistent with FIG. 12.

[0048] At act 2008, the substrate is etched to form a trench or recess in the active region. In some embodiments, act 2008 can be consistent with FIG. 13.

[0049] At act 2010, a germanium layer is epitaxially grown to partially or entirely fill the trench or recess. In some embodiments, act 2010 can be consistent with FIGS. 14A-14B.

[0050] At act 2012, which is optional, a silicon germanium layer is formed to overlie the germanium layer in situ in the same cluster tool in which the germanium layer was grown. In some embodiments, act 2012 can be consistent with FIGS. 15A-15B. Box 2011 indicates these acts may be performed in-situ in the same cluster tool.

[0051] At act 2014, a silicon cap layer is grown to overlie the germanium layer and/or overlie the silicon germanium layer in situ in the same cluster tool in which the germanium layer was grown. In some embodiments, act 2014 can be consistent with FIGS. 16A-16C.

[0052] At act 2016, a photodiode is formed by implanting dopants into the germanium layer. In some embodiments, act 2016 can be consistent with FIG. 17.

[0053] At act 2018, a protective oxide layer is formed over the silicon cap layer. In some embodiments, act 2018 can be consistent with FIG. 18.

[0054] At act 2020, contacts to the photodiode are formed. In some embodiments, act 2020 can be consistent with FIG. 19.

[0055] In some embodiments, the present disclosure relates to an optoelectronic device, comprising: a substrate, and a germanium photodiode region extending into an upper surface of the substrate. The germanium photodiode region has a curved upper surface that extends past the upper surface of the substrate. A silicon cap overlies the curved upper surface of the germanium photodiode region, and there is an absence of oxide between the curved upper surface of the germanium photodiode region and an upper surface of the silicon cap.

[0056] Other embodiments relate to a method. In the method, a trench is formed in a substrate, and the substrate, including the trench, is placed within a chamber. Within the chamber, a germanium layer is epitaxially grown in the trench to have a curved upper surface, and in situ within the chamber, a silicon cap layer is epitaxially grown over the germanium layer.

[0057] Still other embodiments relate to a method. In this method, a substrate is received, and a shallow trench isolation region is formed to surround an active region in the substrate. A doped region is formed in the substrate, and an oxide layer is formed over an upper surface of the substrate and over the shallow trench isolation region. The substrate is etched to form a trench or recess in the active region, and a germanium layer is epitaxially grown to partially or

entirely fill the trench or recess. A silicon cap layer is then epitaxially formed to overlie the germanium layer. The silicon cap layer is formed in situ in a cluster tool in which the germanium layer was grown without exposing the germanium layer to an ambient environment surrounding the cluster tool.

[0058] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An optoelectronic device, comprising:
 - a substrate;
 - a photodiode structure comprising a first semiconductor material and extending into an upper surface of the substrate, wherein the photodiode structure has a curved upper surface that extends past the upper surface of the substrate; and
 - a cap structure overlying the curved upper surface of the photodiode structure, wherein the cap structure comprises a second semiconductor material different from the first semiconductor material and wherein there is an absence of oxide between the curved upper surface of the photodiode structure and an lower surface of the cap structure, and wherein the cap structure directly contacts the entirety of the curved upper surface of the photodiode structure.
2. The optoelectronic device according to claim 1, further comprising:
 - an interface layer disposed between the photodiode structure and the cap structure, the interface layer including a first atomic species of the first semiconductor material and including a second atomic species of the second semiconductor material.
3. The optoelectronic device according to claim 1, wherein the first semiconductor material of the photodiode structure comprises germanium and the second semiconductor material of the cap structure comprises silicon.
4. The optoelectronic device according to claim 2, wherein the first semiconductor material of the photodiode structure comprises germanium, the second semiconductor material of the cap structure comprises silicon, and the interface layer comprises silicon germanium.
5. The optoelectronic device according to claim 2, wherein the interface layer has a constant atomic ratio of the first atomic species to the second atomic species throughout a thickness of the interface layer.
6. The optoelectronic device according to claim 2, wherein the interface layer has a changing atomic ratio of the first atomic species to the second atomic species over a thickness of the layer.
7. The optoelectronic device according to claim 6, wherein the changing atomic ratio has a first value nearer the photodiode structure and a second value nearer the cap structure, the second value being greater than the first value.

8. The optoelectronic device according to claim **1**, further comprising:

- a protective dielectric structure disposed over the cap structure, the protective dielectric structure having a curved upper surface over the cap structure.

9. The optoelectronic device of claim **8**, wherein the protective dielectric structure has a central region over the photodiode structure and has outer edges that extend laterally over the upper surface of the substrate past outer sidewalls of the photodiode structure.

10. The optoelectronic device of claim **1**, further comprising:

- an isolation structure comprising dielectric extending into the upper surface of the substrate on opposite sides of the photodiode structure.

11. The optoelectronic device of claim **10**, wherein an inner region of the substrate separates an inner edge of the isolation structure from an outer sidewall of the photodiode structure.

12. The optoelectronic device of claim **10**, further comprising:

- a p-type region to a first side of the photodiode structure and at a first location laterally beyond a first outer sidewall of the isolation structure furthest from the photodiode structure; and

- an n-type region to a second side of the photodiode structure and at a second location laterally beyond a second outer sidewall of the isolation structure furthest from the photodiode structure; and

- wherein the p-type region, the n-type region, and the photodiode structure establish a p-n photodiode or a p-i-n photodiode.

13. An integrated circuit, comprising:

- a substrate;

- a germanium structure extending into an upper surface of the substrate, wherein the germanium structure has a curved upper surface that extends past the upper surface of the substrate;

- an oxide layer disposed over the upper surface of the substrate and disposed over a peripheral region of the curved upper surface of the germanium structure;

- a silicon germanium interface structure disposed over a central region of the curved upper surface of the germanium structure and having an outer sidewall adjacent to an inner sidewall of the oxide layer;

- a silicon cap layer disposed over a portion of the oxide layer over the peripheral region of the curved upper surface of the germanium structure and extending over the silicon germanium interface structure.

14. The integrated circuit of claim **13**, wherein the oxide layer has a lower portion that extends into the upper surface of the substrate on opposite sides of the germanium structure.

15. The integrated circuit of claim **13**, further comprising:

- a protective dielectric structure disposed over the silicon cap layer, the protective dielectric structure having a curved upper surface over the silicon cap layer.

16. An optoelectronic device, comprising:

- a substrate;

- a photodiode structure extending into an upper surface of the substrate, wherein the photodiode structure has a curved upper surface that extends past the upper surface of the substrate, the photodiode structure comprising a first semiconductor material;

- a cap structure over the curved upper surface of the photodiode structure, wherein there is an absence of oxide between the curved upper surface of the photodiode structure and a lower surface of the cap structure and wherein the cap structure comprises a second semiconductor material that differs from the first semiconductor material;

- a first doped region to a first side of the photodiode structure, the first doped region having a first conductivity type; and

- a second doped region to a second side of the photodiode structure, the second doped region having a second conductivity type; and

- wherein the first doped region, the second doped region, and the photodiode structure establish a p-n photodiode or a p-i-n photodiode.

17. The optoelectronic device of claim **16**, wherein the photodiode structure extends along a linear segment as viewed from above the upper surface of the substrate.

18. The optoelectronic device of claim **17**, wherein the photodiode structure comprises a rounded upper surface that extends axially over the linear segment.

19. The optoelectronic device of claim **18**, wherein the photodiode structure comprises a body and distal ends arranged along the linear segment, wherein at least one of the distal ends of the photodiode structure contacts a silicon core that extends into the upper surface of the substrate.

20. The optoelectronic device of claim **19**, wherein the at least one of the distal ends where the photodiode structure contacts the silicon core has a tapered shape, such that the at least one of the distal ends is narrower than the body of the photodiode structure.

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